

EXPERIMENT 11 - STUDY OF GPIO & BIT ADDRESSING IN AT89C51 USING EMBEDDED C

Registers

Register	Value
r0	0x00
r1	0x00
r2	0x00
r3	0x00
r4	0x00
r5	0x00
r6	0x00
r7	0x00
SP	0x07
DPTR	0x0000
PC	C:0x0821
STATUS	391
SEC	0.000195...
PSW	0x00

Disassembly

```
25: delay();
26:
27: LCALL delay(C:0800)
28: LED1 = 0; // LED1 OFF
```

embedded11.c

```
11: for(i = 0; i < 200; i++)
12: {
13:     for(j = 0; j < 1000; j++);
14: }
15: // Main function
16: void main()
17: {
18:     while(1)
19:     {
20:         LED1 = 1; // LED1 ON
21:         LED2 = 0; // LED2 OFF
22:         delay();
23:         LED1 = 0; // LED1 OFF
24:         LED2 = 1; // LED2 ON
25:         delay();
26:     }
27: }
```

Parallel Port 1

Port 1	7 Bits	0
P1: 0xFE	✓✓✓✓✓✓✓	✓
Pins: 0xFE	✓✓✓✓✓✓✓	✓

Command

```
Running with Code Size Limit: 2K
Load "D:\CS1\Examples\HELLO\Objects\embedded11"
```

Call Stack + Locals

Name	Location/Value	Type
MAIN	C:0x081D	

Simulation

t1: 0.00019550 sec L:13 C:6 CAP NUM SCRL OVR: R/W

30°C Sunny

Registers

Register	Value
r0	0x00
r1	0x00
r2	0x00
r3	0x00
r4	0x00
r5	0x00
r6	0x00
r7	0x00
SP	0x07
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PC	C:0x0821
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Disassembly

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25: delay();
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23:         LED1 = 0; // LED1 OFF
24:         LED2 = 1; // LED2 ON
25:         delay();
26:     }
27: }
```

Parallel Port 1

Port 1	7 Bits	0
P1: 0xFD	✓✓✓✓✓✓✓	✓
Pins: 0xFD	✓✓✓✓✓✓✓	✓

Command

```
Running with Code Size Limit: 2K
Load "D:\CS1\Examples\HELLO\Objects\embedded11"
```

Call Stack + Locals

Name	Location/Value	Type
MAIN	C:0x081D	

Simulation

t1: 0.00019550 sec L:13 C:6 CAP NUM SCRL OVR: R/W

30°C Sunny